

17.1

Kubernetes Architecture

Control Plane, Worker Nodes & Core Components

Java Backend Architecture Interview Series • Tutorial Edition • Easy Diagrams & Examples

What is Kubernetes?

Kubernetes (K8s) is a platform that automatically runs and manages your containers. Think of it as an 'operating system for your cluster' — you tell it WHAT to run and HOW MANY copies, and Kubernetes figures out WHERE to run them, restarts them if they crash, and scales them up/down based on load.

■ **Simple Analogy:** Imagine a shipping company with hundreds of trucks (nodes). A central dispatch office (control plane) decides which truck carries which cargo (pod). If a truck breaks down, dispatch immediately reroutes its cargo to another truck.

Two Main Parts of Kubernetes

Control Plane	The 'brain' of Kubernetes. Runs on master nodes. Makes all decisions about the cluster.
Worker Nodes	The 'muscle'. Run your actual application containers (Pods).
Pod	The smallest unit K8s manages. Usually 1 container, shares network + storage.
kubectl	The command-line tool you use to talk to Kubernetes.

Diagram — Kubernetes Cluster Architecture

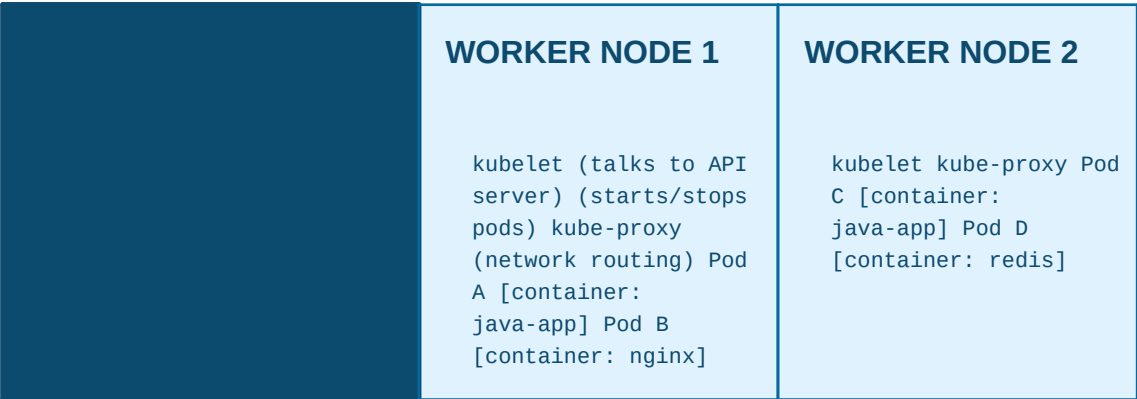


Figure 1: Control Plane manages the cluster; Worker Nodes run your Pods

Control Plane Components Explained

Component	What It Does	Analogy
kube-apiserver	Front door — validates and saves all requests	Reception desk
etcd	The cluster's database — stores every resource as key-value	Company database
kube-scheduler	Picks which node each new Pod runs on	HR assigning offices

kube-controller-manager	Watches state, fixes drift — e.g., restarts crashed pods	Help desk
cloud-controller-manager	Connects K8s to AWS/GCP/Azure APIs	Cloud liaison

Quick kubectl Cheat Sheet

```
kubectl get nodes # list all nodes
kubectl get pods -n my-namespace # list pods in namespace
kubectl describe pod my-pod # full pod details
kubectl logs my-pod -f # stream logs
kubectl exec -it my-pod -- bash # shell into pod
kubectl apply -f deployment.yaml # deploy from file
kubectl rollout status deploy/my-app # watch rolling update
```

■ **Real-World Example:** Google runs Kubernetes internally as 'Borg' — it managed 2+ billion container starts per week. The open-source K8s was donated by Google in 2014. Today GKE, EKS, and AKS manage the control plane for you — you just add worker nodes.